

Xiangbo Gao

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EDUCATION

University of California, Irvine (GPA-3.720) 2018.9 - Present
Double Majors: **Computer Science, Mathematics** BS expected 2023.3
ICS Honor Program, Dean's List
University of California, Berkeley, Summer Session (GPA-3.566) 2019.6- 2019.9
Core Courses: Python (CS 61A), Data Structure (CS 61B), Intro to AI (CS188)

PUBLICATIONS

1. **X Gao**¹, et al. "Scale-free and Task-agnostic Attack: Generating Photo-realistic Adversarial Patterns with Patch Quilting Generator" arXiv:2208.06222, 2022 (On Submission)
2. M Liu, X Li, **X Gao**³, J Chen, L Shen, & H Wu "Sample hardness based gradient loss for long-tailed cervical cell detection", International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 109-119. Springer, Cham, 2022

RESEARCH EXPERIENCE & LEADERSHIP

Imperceptible Adversarial Attack via Generative Model 2022.11 ~ Present
PI: Prof. WeiCheng Xie | Inst: Institute of Computer Vision, Shenzhen University, China

- Use Generative networks to generate adversarial perturbation on the human imperceptible areas of images in an end-to-end manner.

Auto-generated Graphical Model in the Autonomous Driving System 2022.5 ~ Present
PI: Prof. Mohammad Al Faruque | Inst: University of California, Irvine, CA

- Designed and created a multi-domain autonomous driving dataset for various driving scenarios using the CARLA simulator.
- Developed a probabilistic LSTM structure that encodes states as variational embeddings.
- Applied knowledge from PointNet to encode unordered lane mark information.
- Evaluating and comparing the model performance and cross-domain transferability with other motion prediction algorithms, such as MTP, PGP, and Trajectron++.

Multi-modal 3D Object Detection in autonomous driving scenario 2022.6 ~ Present
PI: Xiangbo Gao | Inst: University of California, Irvine, CA

- Conducted a literature review of existing general 3D object detection algorithms, including camera-based, point-based, voxel-based, and multi-modal algorithms.
- Ran experiments on multimodal 3D object detection algorithms such as Lidar Painting and Bev fusion.

Goal-conditional Reinforcement Learning 2022.2 ~ 2022.6
PI: Prof. Roy Fox | Inst: University of California, Irvine, CA

- Conducted a literature review on imitation learning and general reinforcement learning.
- Proposed a Spring Loss that uses the idea of contrastive learning to align the embeddings in the linear distance. Visualized the learned embedding using PCA, which shows better alignment with real-world trajectories.

Adversarial Attack with Semantic Pattern

2021.4 ~ 2022.10

PI: Prof. WeiCheng Xie and Siyang Song | Inst: Institute of Computer Vision, Shenzhen University, China

- Proposed a novel Patch Quilting Generative Adversarial Network (PQ-GAN) that learned a set of cascaded generators to generate image patterns of varying scales without distortion.
- Applied the PQ-GAN to adversarial attacks that delivered state-of-the-art attack strength and robustness with respect to various types of defense algorithms

Long-tailed Cervical Cell Detection

2021.5 ~ 2022.1

PI: Prof. Linlin Shen | Inst: Institute of Computer Vision, Shenzhen University, China

- Conducted experiments and contributed to 20% of paper writing for a proposed Grad-Libra Loss that uses gradients to dynamically calibrate sample hardness and rebalance gradients.

ZerO Waste Anteaters

2020.9 ~ 2022.4

PI: Prof. Sharad Mehrotra | Inst: University of California, Irvine, CA

- Led a team of up to 8 members to explore waste recognition solutions.
- Trained and deployed lightweight waste image classification and object detection models.
- Designed a small Image Recognition tutorial with accompanying side projects to help interested students get started in this field.

INTERNS & EMPLOYMENTS

Chinese Union in Computer Science, University of California, Irvine

2020.4 ~ 2021.3

Position: Member of CUCS Academic Department

Tandll Investment Management Limited, Shenzhen, China

2020.6 ~ 2020.8

Position: Full-stack Software developer

Calit 2, University of California, Irvine

2019.2 ~ 2019.7

Position: Software developer

COMPETITIONS

Netease Hackathon Competition

2020.6

Outstanding Award: A website for real-time traffic statistical monitoring using React, GraphQL, Rest, and MongoDB

Competition, University of California, Irvine, CA, USA

2020.4

1st place in Machine Learning Hackathon at UCI

Geometric Pose Affordance: predicting 3D pose from 2D data and light tracing information

Google Hash Code 2020 Algorithms Competition, Irvine, CA

2020.2

Result: 2nd place / 13 at UCI

ACTIVITIES

UCI Rock Climbing Group

SKILLS

Programming Language: Python, C/C++, Java, C#, JavaScript, SQL

Software: PyTorch, OpenMMLab, RLib, TorchGeometry, TensorFlow